WebClient Computing Using jQuery

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How to Use jQuery

- Add a reference to the minified Javascript file
- Adding jQuery to a page using CDN
  - Google jquery CDN and search for the link for 2.x version
    - http://code.jquery.com/jquery-2.2.3.min.js
  - In the code, add as a general js file

```
<head>
  <title>jQuery exercise</title>
  <script type="text/javascript" src="http://code.jquery.com/jquery-2.2.3.min.js"></script>
</head>
```

Access jQuery Object

- $ sign
  - A magic character to access object in jQuery
  - Returns a root jQuery object
- $(item_selection)
  - Find an item or group of items
  - Register event handler for the corresponding content

```
<body>
  <div id="target">Demo Element</div>
</body>
```

```
> $('#target')
<· [  <div id="target">Demo Element</div>]
```

Modifying Contents (Text) Using jQuery

- jquery_obj.html(‘html_contents_to_replace’);
  - Gets or replaces the existing HTML in an element
- jquery_obj.text(‘text_value_of_element’);
  - Gets or replace the existing text in an element

```
$('#target').text('updated content');
$('#target').html('HTML updated content');
$('#target').html('<b>HTML</b> updated content');
$('#target').text('<b>text</b> updated content');
```

jquery_value.js
jquery_class.html

Modifying Attributes Using jQuery

- Reading an attribute value
  - $(element_identifier).attr(’attribute-name’);
- Writing a attribute value
  - $(element_identifier).attr(‘attribute-name’, ‘value’);
- Remove an attribute
  - $(element_identifier).removeAttr(‘attribute-name’);
- Access css property
  - $(element_identifier).css(‘css attribute name’, ‘optional value to be set’)

```
$('#target').attr('style', 'color:red; font-size:3em;');
var attr_style = $('#target').attr('style');
var css_color = $('#target').css('color');
$('#display').text("style attribute: " + attr_style + " css color: " + css_color);
$('#target').css('color', 'blue');
$('#target').removeAttr('style');
```

jquery_attribute.js
jquery_class.html

Referencing CSS Selector or ID Using jQuery

- Using $(SelectorName)
- What happens multiple items with a same selector name
  - In Javascript, an array of multiple elements were returned
  - In jQuery, all the elements take the effect
- Changing text of an element
  - $(‘h1’).text(”change all contents of h1”);

jQuery Exercise

- jquery.html and jquery_selector.js

```html
<!DOCTYPE html>
<html>
<head>
  <title>jQuery exercise</title>
  <script type="text/javascript" src="http://code.jquery.com/jquery-2.2.3.min.js"></script>
</head>
<body>
  <h1>This is h1</h1>
  <h1 class="h1-class">h1 with a class name</h1>
  <h1 id="h1-id">h1 with an ID</h1>
</body>
</html>
```

```javascript
$('h1').text() = "Hello jQuery";
$('h1.h1-class').text() = "Hello jQuery class";
$('h1#h1-id').text() = "Hello jQuery ID";
$('#h1-id').text() = "ID without tag name";
$('.h1-class').text() = "class without tag name";
```

Compare No jQuery – Using Plain Javascript

```
> var t = document.getElementById("first-p");
<· undefined
> t
<· ▼<p id="first-p">
      "The "
      <strong>first</strong>
      " paragraph"
    </p>
> t.innerHTML
<· "The <strong>first</strong> paragraph"
> t.innerText
<· "The first paragraph"
```

```
> t.innerText = "Changing content";
<· "Changing content"
> t.style.fontSize = "3em";
<· "3em"
> t.style.color="red";
<· "red"
> t=document.getElementsByClassName("class-1");
<· [  <p class="class-1">Class 1 paragraph</p>,
    <div class="class-1">Class 1 div</div>]
> t.length
<· 2
> t[0]
<·   <p class="class-1">Class 1 paragraph</p>
> t[1]
<·   <div class="class-1">Class 1 div</div>
```

Traversing DOM Tree Using jQuery

- jQuery provides method to traverse DOM tree based on a current element
  - Child elements iteration
  - Going to parent node
  - Traversing sibling nodes

Walk Down the Tree

- children(optionalSelector) – only get direct children nodes
  - children(“.class_name”);
  - children(“#id_name”);
- find(optionalSelector) – get any descendant nodes
  - find(“.class_name”);
  - find(“#id_name”);

Walking Up the Tree

- parent(conditional statement) : direct parent only
  - parent(“.class_name”);
  - parent(“#id_name”);
- parents(conditional statement) : get all the ascendant nodes
  - parents(“.class_name”);
  - parents(“#id_name”);

Parent-Child Examples

- jquery_parent_child.html and jquery_parent_child.js

```
var start = $('#container');
start.children('#second-child').text("this is my second child");
start.find('.second-gc').text('onlt the second grand child');

var fgc = $('#first-gc');
fgc.parent().text("You are my parent");
fgc.parents("#container").text("Changing container");
```

```
<body>
  <div id = "container">Top Most div
    <div id="first-child">First child
      <p id="first-gc">First grandchild</p>
      <p class="second-gc">Second grandchild</p>
    </div>
    <div id="second-child">Second child</div>
  </div>
</body>
```

Reference

- edx jQuery course
- w3schools.com
- http://api.jquery.com/